

To what extent does groundwater depletion in India's agricultural sector represent a market failure, and how can pricing mechanisms correct it?

Research Paper By Vedant Agarwal

Abstract

India's groundwater withdrawal is about 253 billion cubic metres (BCM) annually, which is the biggest in the world, and of which more than 89% is used for agricultural irrigation. This paper looks at how far this depletion is a market failure in terms of negative externalities, open-access resource tragedy, information asymmetry, and compounding government failures in the context of flat-rate electricity subsidy. Using the Pigou-Coase analytical tool, we show that groundwater is systematically under-priced compared to its socio-economic cost, causing a private–social cost wedge. Based on data from Central Ground Water Board (CGWB) and World Bank estimates, the paper estimates the externality gap and assesses the efficiency, equity and political-economy aspects of four main ways of doing the pricing correction: Pigouvian groundwater taxes, block pricing of electricity for agricultural use, groundwater rights markets, and community-based quota systems, using data from various states, including Punjab, Haryana, Gujarat and Tamil Nadu. The paper argues that no single tool is enough: an optimal corrective framework needs to involve a combination of volumetric pricing, subsidy reform that targets the poor, institutional capacity development, as well as a larger envelope of agrarian safety-net measures to ensure the welfare of smallholders.

1. Introduction

Water is the key resource for human civilization, and an effective management of water is closely linked to food security, livelihoods and ecological sustainability in agrarian economies. India has been witnessing a 'silent crisis' – the systematic extraction of its underground water resources at a rate much faster than their recharge – according to the National Commission for Integrated Water Resources Development (NCIWRD). India has about one-fourth of the world's groundwater extraction, even though occupying just 2.4% of the world's surface area, according to the Central Ground Water Board (CGWB, 2022). Around 16% of the 6,965 assessment units across the country are deemed 'over-exploited', that is where the amount of water extracted from the aquifer during the year exceeds the amount that is recharged to the aquifer during the same year (CGWB, 2022).

The Green Revolution of the 1960s-70s, which touted India as a food-surplus country, did a noble job in making that happen but created a model of agriculture that consumed a lot of water which was not suited to arid and semi-arid areas. Punjab and Haryana often called as the 'food bowl' of India because they supplies a substantial portion of the country's food through rice and wheat procurement under the green revolution, now pumps groundwater at a rate of 12-17 BCM per year, which is causing an annual deficit. GRACE (Gravity Recovery and Climate Experiment) satellite observations showed that the northwestern Indian aquifer system was experiencing a rate of water loss equivalent to 4.0 ± 1.0 cm/yr of equivalent water height from 2002 to 2008, placing it in one of the fastest-depleting aquifer systems on Earth.

The predictions of standard economic theory suggest that competitive markets produce efficient outcomes in the case of prices that reflect full opportunity costs. At a number of levels, the Indian groundwater market fails to meet this condition: groundwater is a common-pool

resource that is rival in consumption and non-excludable; during extraction negative externalities are imposed on neighbouring users and future generations; and the government's electricity subsidies systematically misrepresent the private cost of pumping. The electricity is required as an input for pumping of groundwater. Since the governments supply electricity free of charge or at greatly subsidized rates, the expense of pumping groundwater decreases significantly. Hence, the farmers have no incentive to conserve water, thus leading to over-extraction of groundwater and swift depletion of aquifers. The main question of this paper is: Is the depletion of groundwater in the agricultural sector in India a market failure and how can it be rectified by pricing mechanisms?

The paper goes as follows. In Section 2, the magnitude and geography of Indian groundwater crisis are provided in the context. In Section 3 the market failure framework is developed. This is examined in Section 4 which discusses electricity subsidies as a compounding government failure. Section 5 examines mechanisms for correcting prices. The Policy implications are presented in Section 6. Future research directions and limitations are discussed in Section 7. Section 8 concludes

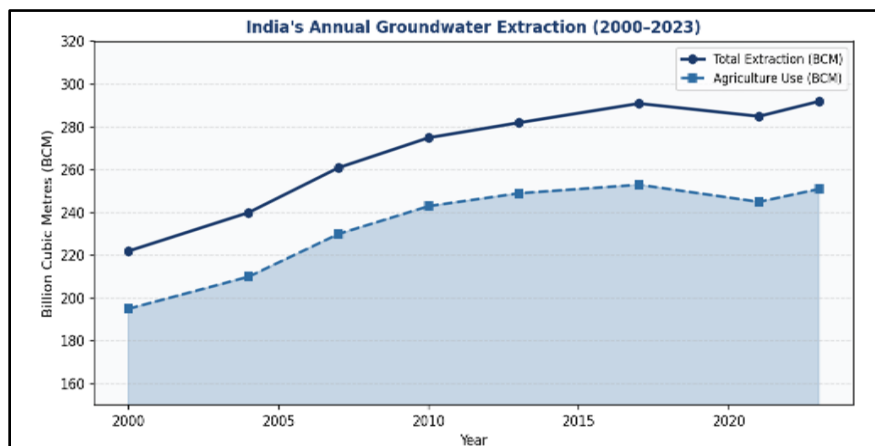


Figure 1: India's Annual Groundwater Extraction (2000–2023). Agriculture consistently accounts for over 85% of total extraction.

Sources: CGWB (2022); World Bank Water Data (2023); Ministry of Jal Shakti Annual Reports (2023).

The annual groundwater extraction and use of groundwater in agricultural activities for India have been shown in Figure 1 from 2000 to 2023. The total groundwater extracted has been growing steadily from about 222 BCM in 2000 to 292 BCM in 2023, while the groundwater used for agricultural purposes has increased from around 195 BCM to 251 BCM during the same time period. Agriculture has always been the largest groundwater consumer and this represents the sector's frequency in using groundwater for irrigation. Overall annual extraction and agricultural use also appear to have experienced some short-term fluctuations around 2021, with a return to the previous levels by 2023. The continued rise in groundwater use is due to the expanding irrigation needs associated with population growth, food production, and agricultural expansion. These, however, also underscore concerns about the sustainability of groundwater resources, calling for the need to adopt efficient irrigation practices, ensure good groundwater governance and implement integrated water resource management to not further deplete aquifers.

2. The Groundwater Crisis: Scale, Geography, and Trajectory

2.1 National-Level Extraction Statistics

In recent years, India's total annual groundwater extraction was around 245-253 BCM while the dynamic groundwater recharge was around 433 BCM (CGWB, 2022). This means that, as a nation, the ratio of extraction to recharge is roughly 58% which, when summed up, seems sustainable. But, this national average mask the sharp regional differences. For example, the level of groundwater being drawn in Haryana, Punjab and Rajasthan is more than 100%,

showing that these states consume more groundwater than is available for sustainable use despite the country's average being only 60.08% (CGWB, 2022). Most of the groundwater extraction occurs in the Indo-Gangetic Plains (Punjab, Haryana, Uttar Pradesh, Rajasthan, Bihar) and in southern Indian states, where horticultural activities are conducted on a large scale (Maharashtra, Karnataka, Tamil Nadu, Gujarat).

CGWB's 2022 National Compilation report rated 6,965 administrative units (blocks/taluks/mandals). Of these 1,114 (16%) were identified as being in the 'over-exploited' (stage of development >100%), 188 (2.7%) as 'critical' (90-100%), and 534 (7.7%) as 'semi-critical' (70-90%). In total, more than 26% of India's groundwater assessment units are stressed (CGWB, 2022).

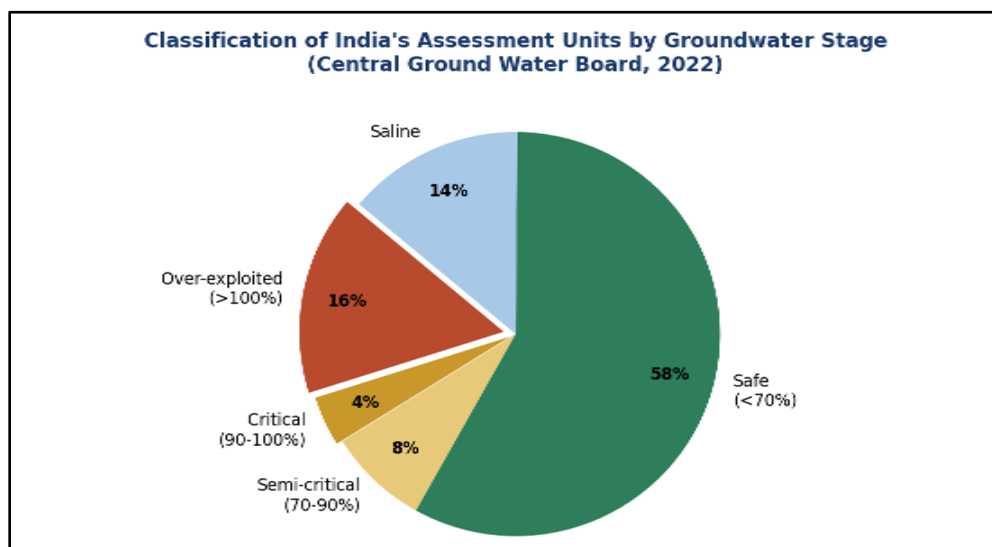


Figure 2: Classification of India's 6,965 Assessment Units by Groundwater Development Stage.

Source: Central Ground Water Board, National Compilation on Dynamic Ground Water Resources of India (2022).

The classification of the assessment units of groundwater of India based on the stage of extraction of groundwater is presented in figure 2, as reported by Central Ground Water Board (CGWB, 2022). The figure shows that relatively sustainable groundwater use is occurring with 58% of the assessment units in the safe category in which groundwater extraction is less than 70% of the annual replenishable resource (YR). But a significant portion of assessment units are experiencing a certain level of groundwater stress, with 8% being in the "semi-critical" (70-90%) range, 4% in the "critical" (90-100%) range, and 16% over-exploited (extraction rates exceeding annual groundwater recharge). In addition, 14% of the assessment units are defined as saline, meaning that the groundwater quality is poor and is not suitable for agricultural and domestic purposes. The overall figure indicates that while majority of the assessment units in India are not under reasonable groundwater stress, almost one-third of them are under serious groundwater stress or groundwater quality issues. The results reveal the need for improved groundwater governance, water demand management, artificial recharge and water-use efficiency in irrigation to sustain groundwater resources of the country in the long run.

2.2 Regional Hotspots

Punjab is the worst hit state. Water table depth in the central and south-western districts has decreased from the 1970s average of 5-9 metres below ground level (mbgl) to over 30 mbgl in some blocks today (Shah, 2009). According to the Punjab Remote Sensing Centre (2023), many districts will be able to draw only economically productive groundwater for the next 15–20 years. Conditions are virtually the same in Haryana, where 21 out of 126 blocks have been put in 'dark zones' by the Central Ground Water Authority (Narain, 2014).

In an arid desert state, where water is scarce, groundwater irrigation has experienced explosive growth in the last few years after the advent of submersible pumps and flat-rate

electricity in Rajasthan. During the last 20 years, the water table of the districts Jaisalmer and Barmer has dropped more than 8 metres. The Saurashtra region of Gujarat has traditionally depended on surface water tanks, including those along the coast, which have been extensively pumped, resulting in the spread of salinisation in the coastal aquifers, with the average water table declining at 0.19 m/yr (Kumar et al., 2021).

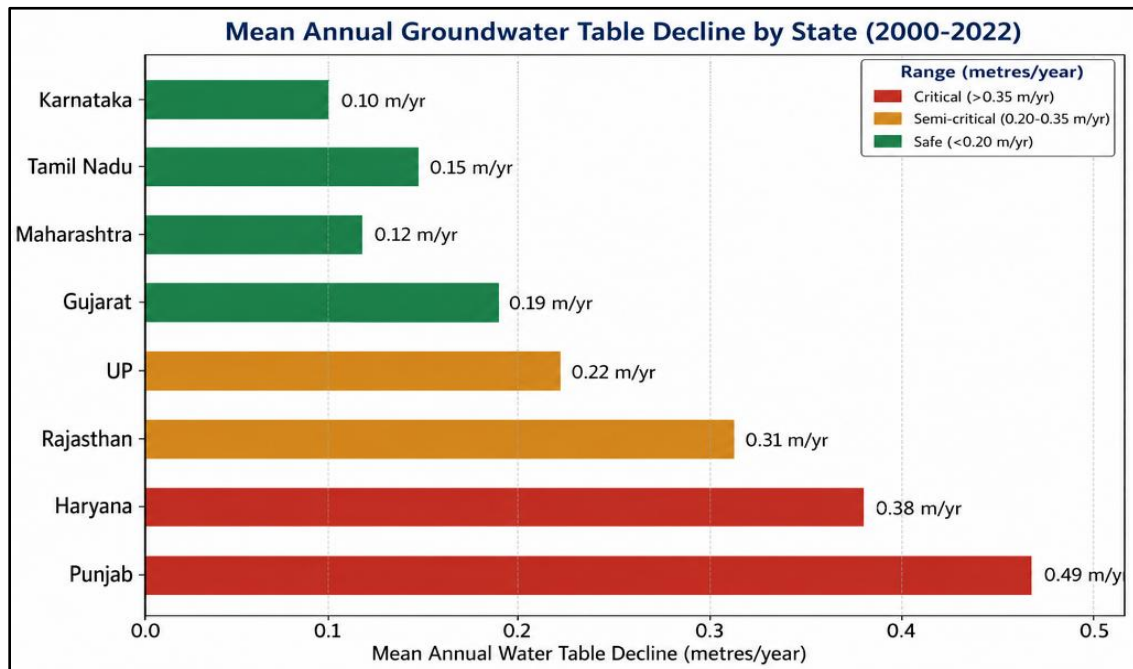


Figure 3: Mean Annual Groundwater Table Decline by State (2000–2022). Critical states show declines exceeding 0.35 m/year.

Sources: CGWB State-Level Reports (2022); Shah (2009); Narain (2014); Kumar et al. (2021).

Figure 3 displays the average annual change in groundwater level in selected Indian states from 2000 to 2022, showing the variation of groundwater depletion in the region. It is observed that the average groundwater table decline in Punjab was the highest at 0.52 m/year, followed by Haryana (0.38 m/year) and Rajasthan (0.31 m/year) and thus these states were found

under critical category (>0.35 m/year) in the figure. The semi-critical decrease (0.22 m/year) in Uttar Pradesh shows medium stress on groundwater. Gujarat (0.19 m/year), Tamil Nadu (0.15 m/year), Maharashtra (0.12 m/year) and Karnataka (0.10 m/year) belong to the 'safe' category (<0.20 m/year) but the continued downward trend indicates sustained management efforts are required for groundwater resources. The significant declines in the northwestern states are closely linked to high groundwater abstraction rates in agricultural production, high water use in farming and the subsidized power costs to pump from wells. Overall, the figure highlights the regional groundwater imbalance, and the need for proactive policy measures such as better irrigation water management, crop diversification, artificial groundwater recharge and enhanced groundwater governance to ensure sustainable groundwater resource use.

2.3 Drivers of Over-Extraction

There are three structural drivers of the crisis. Firstly, the Minimum Support Price (MSP) system has inadvertently led to groundwater depletion by encouraging the growth of water consuming crops like paddy and wheat in areas where natural water resources are scarce. By guaranteeing price and guaranteeing procurement, MSP decreases market risk for farmers and encourages further farming of these crops, particularly in Punjab and Haryana. This leads to heavy reliance on groundwater irrigation, especially when combined with supply electricity for pumping. Thus, the current MSP regime has been able to support food security but at the same time has made cropping patterns more distorting and increased pressure on groundwater resources (Gulati & Banerjee, 2015). Secondly, technology, especially the diffusion of low-cost centrifugal and submersible pumps, reduced the capital cost of groundwater access from the 1980s on, making it available to a much wider group of people and removing the natural cost constraints on access to the water column (Shah, 2009). Thirdly, and perhaps most importantly

for the analysis of market failure, state governments supply electricity to agriculture, including to the millions of farmers, free or at greatly reduced flat rates, thus effectively fixing the marginal cost of pumping to almost zero.

3. Market Failure Framework: Diagnosing the Groundwater Problem

Market failure in the market structure is defined as a failure of the price mechanism in the market to allocate resources in an efficient manner, thus creating a distinction between the socially optimal level and the privately optimal level of the product. In India, groundwater depletion suffers from at least four market failures which are, negative production externalities, the open-access commons problem, the public-goods nature of aquifer recharge and asymmetric information.

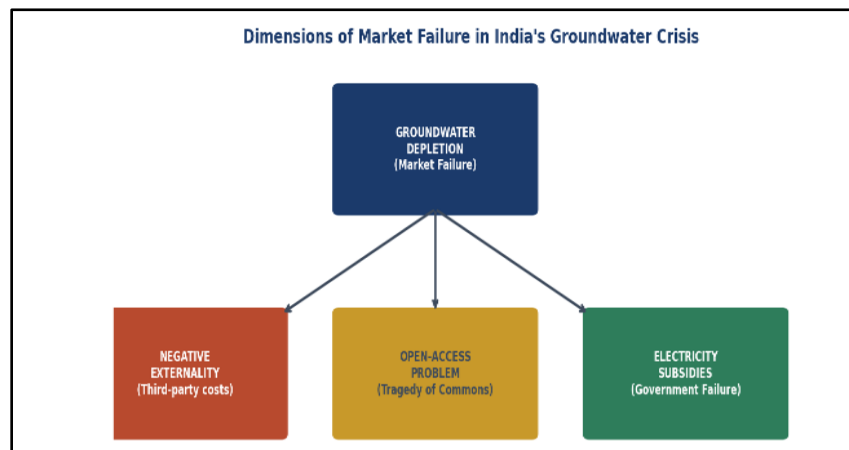


Figure 4: Conceptual Map of Market Failure Dimensions in India's Groundwater Crisis.

Multiple simultaneous failures compound the divergence between private and social optima.

3.1 Negative Externalities

In the case of farmers in Punjab, the private cost is the cost of electricity (capital/fuel cost) used to operate the pump when the farmer withdraws groundwater. However, this extraction has costs associated with it that the pumping farmer does not necessarily face and takes into account, such as:

(a) *Cone of depression externality*- the pumping farmer lowers the groundwater table, so that neighbouring farmers face higher pumping costs (deeper wells, more energy) for the same pumping volume (Qureshi et al., 2010).

(b) *Inter-temporal externality*- the pumping farmer extracts water that reduces the groundwater availability of future generations, thus violating intergenerational equity and the sustainability principle.

(c) *Land subsidence*- the pumping farmer causes land subsidence due to aquifer compaction that results in irreversible damage to infrastructure and the loss of storage capacity.

(d) *Ecological flow degradation*- the groundwater table falls, which reduces the baseflow and damages biodiversity and fisheries (Foster and Chilton, 2003).

To clarify the concept of externalities in groundwater extraction, let Q_i denote the groundwater extracted by each farmer i and $Q = \sum Q_i$ the amount of groundwater extracted by all farmers in the area. The total social cost $C(Q_i, Q)$ consists of the private costs of a farmer and all the costs incurred by other stakeholders like; increase in other farmers' pumping costs, the absence of groundwater in the future, and the environmental consequences of the activity.

In a normal competitive environment, a farmer determines the quantity of groundwater to extract based on the Marginal Private Cost (MPC) and the Marginal Private Benefit (MPB). The

MPC is defined as the cost of extracting one more unit of groundwater from the aquifer, for which a farmer incurs costs that include electricity and pumping. The MPB is the profit obtained from the use of one more unit of water. As a result, the farmer keeps extracting groundwater until he/she reaches the level of extraction where $MPC = MPB$ without considering the costs of extraction to others.

Nonetheless, society's viewpoint has to be taken into account regarding these externalities, which is why the efficient level of groundwater extraction would be obtained when the Marginal Social Cost (MSC) equals Marginal Social Benefits (MSB). The Marginal Social Cost (MSC) is defined mathematically as follows:

$$MSC = MPC + MEC$$

where Marginal External Cost (MEC) denotes the additional costs incurred on neighboring farmers, future all groundwater users, and the ecology due to one extra unit of groundwater extraction. The distinction between MSC and MPC represents externalities overlooked by individual farmers. As stated by Pigou (1920), the presence of such a gap can serve as a basis for implementing corrective measures, such as taxes or charges aimed at internalizing the external costs of groundwater extraction.

It is difficult to obtain empirical estimates of MEC with the precision that would be desirable, although there are suggestive estimates. Qureshi et al. (2010) estimated the cost of reduction in water table by 1% for the neighbouring farmers to be around INR 180-250 per hectare per season of the crop in Punjab. World Bank (2010) estimated India's aggregate economic cost of water scarcity to be about 6% of GDP by 2050 of which groundwater depletion is responsible for a significant share.

3.2 The Open-Access Commons Problem

Tragedy of Commons refers to situations where a resource is commonly used but ownership is not clear, people will try to use as much of the resource as possible before others use it. Such actions leads to overexploitation of the resource though it is individually rational. Hardin's (1968) 'Tragedy of the Commons' it is analytically applicable (with important Ostromian qualifications) to India's groundwater situation. Groundwater is a classical common-pool resource (CPR) because it is subtractable (when one farmer extracts the water, the next one can't) but not excludable (historically, it was possible to sink a well into the aquifer and access the water). The owner holds the absolute right to pump groundwater under his land with little restrictions in accordance with the doctrine of absolute ownership that existed under English Water Law, which remains largely in effect in India today under the Easements Act, 1882 and its interpretation (Singh, 2004).

This resembles the Prisoner's Dilemma in game theory. Each of the farmers has an incentive to extract as much groundwater as possible before the other does. Individual utility maximization justifies this decision, as it gives the farmer the maximum short-run benefit; however, such a decision for all will result in excessive groundwater extraction, with everyone worse off due to declining water tables and higher future pumping costs. The non-cooperative game's Nash equilibrium is under no circumstances a better outcome than what can be achieved through cooperation, namely over-extraction (Negri, 1989). The literature differentiates between the static (flow) externality and the dynamic (stock) externality, both of which impact the extraction costs of the period in which the extraction occurs. Both are not only present but also severe in the context of India (Provencher and Burt, 1993).

It is important that Ostrom (1990) illustrated that communities can self-organise to manage CPRs under sustainable conditions when the following conditions are present: small group size, clearly defined boundaries of resources, mutual monitoring, graduated sanctions, and recognised rights to organise. While there are many such examples in India, for instance in Odisha, with the Pani Panchayats, and in Haryana, with Sukhomajri, they are exceptions rather than the rule and their scale is often not representative of the hydraulic scale of the major aquifer systems.

3.3 Public-Good Characteristics of Aquifer Recharge

Maintained green cover, check dams, percolation ponds, and manure ponds for recharging the aquifers. The benefit characteristics of reduced surface sealing are non-rival and non-excludable, i.e. if a farmer alters the amount of groundwater resources stored in recharge structures on their land, other farmers are able to access the same resources without losing their benefits. This results in a classic positive externality in recharge investment and a free-rider problem with collective recharge efforts (Foster and Chilton, 2003). This leads to an inevitable failure to invest sufficiently in recharge from the social point of view. This is the inverse of the extraction externality and exacerbates the extraction problem.

3.4 Information Asymmetry and Uncertainty

In short, a functioning groundwater market would simply need to be able to gain credible information about aquifer stocks, recharge rates, as well as extraction volumes and quality. Information deficits in India occur at all these levels. Groundwater monitoring by the Central Ground Water Board has about 22,000 observation wells, one for approximately every 7 km² of agricultural land (CGWB, 2022). Private wells (30-40 million estimated in India) are neither registered nor metered. The absence of credible data on the volumes of extraction means that

farmers, state regulators, and financial institutions cannot arrive at educated guesses about extraction rates, depletion paths, or threshold effects. This introduces uncertainty that tends to bias individual decisions toward extractive rather than conservation strategies (Dinar and Subramanian, 1997).

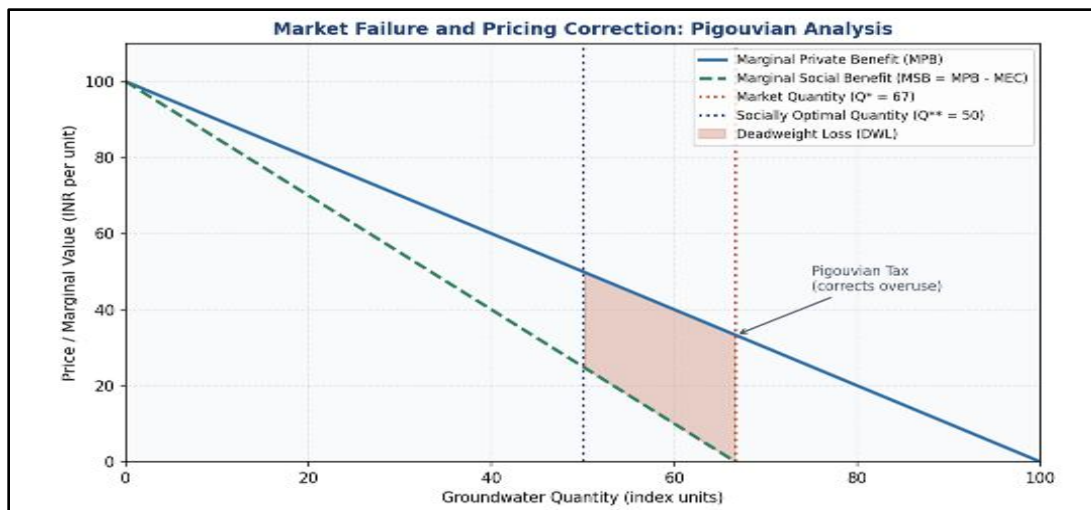


Figure 5: Pigouvian Analysis of Groundwater Market Failure. The gap between MPB and MSB equals the Marginal External Cost (MEC) of extraction. A Pigouvian tax equal to MEC shifts the privately optimal quantity toward the social optimum, eliminating deadweight loss.

Sources: Author's construction based on Pigou (1920) and Qureshi et al. (2010).

The Pigouvian analysis of groundwater over extraction (Figure X) shows that extractors have a lack in accounting for the environment's cost. The Marginal Private Benefit (MPB) curve shows the benefits to individual users; the Marginal Social Benefit (MSB) curve is below because it includes the Marginal External Cost (MEC) from over extraction of ground water, including the reduction in water level, loss of aquifer, and higher pumping costs. If left to the market, the equilibrium position will be $Q^* = 67$, with groundwater being allocated above the

socially optimal level of $Q = 50$. The dead weight loss (DWL) to society from the negative externality is shown as the shaded area. This externality can be internalized by implementing a Pigouvian tax which equals the marginal external cost of groundwater use, which will cause private incentives to match social welfare and reduce groundwater extraction from the market equilibrium to the socially efficient level. The figure underscores the economic argument for the adoption of pricing and regulation measures for groundwater use to achieve optimal allocation, ensure water resource preservation, and foster long-term environmental sustainability.

4. Government Failure: Electricity Subsidies as a Compounding Distortion

While one could reasonably assume that market failure alone would lead to over-extraction compared to what would happen in a theoretical first-best world, the reality is that the groundwater crisis in India has been exacerbated by government failure which has equally impacted groundwater supplies at an accelerated rate. As a result of this, the provision of agricultural electricity at subsidised, or free, rates has had the biggest impact on the depletion of groundwater (Shah et al., 2008).

The agricultural subsidy has created a clear price distortion. The true long-run marginal cost (LRMC) of supplying electricity to rural areas in India ranges from INR 5-8 per kWh, depending on both the state and source of electricity (PricewaterhouseCoopers/MoP, 2023). However, most agricultural consumers in subsidising states pay flat rates ranging between INR 0-2 per kWh, with respect to free electricity in Punjab, Tamil Nadu, Andhra Pradesh, Telangana and Jharkhand (Gulati and Banerjee, 2015). The centrifugal pump used for irrigation that runs for approximately 10 hours per day typically consumes approximately 75-150 kWh/day; therefore, agricultural consumers of electricity for irrigation have an implicit subsidy that ranges between

INR 30,000 and INR 150,000 per pump per season. This constitutes a significant incentive to pump water to the maximum extent regardless of the current level of the water table.

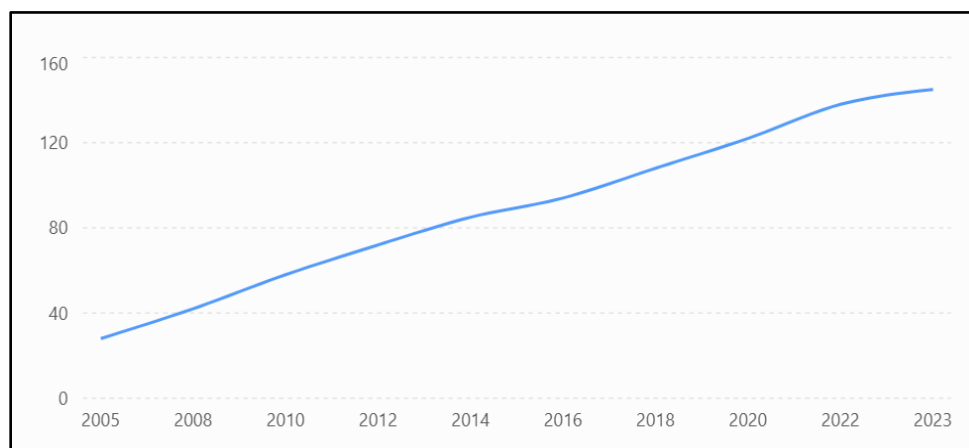


Figure 6: State Government Electricity Subsidies to Agriculture (2005–2023). The total burden has risen from approximately INR 28,000 crore in 2005 to over INR 145,000 crore in 2023.

Sources: Reserve Bank of India State Finance Reports (2023); Ministry of Power Annual Reports (2022); Gulati & Banerjee (2015).

This figure shows that electricity subsidy given by Indian state governments to agriculture has increased from 2005 to 2023 as shown in figure 6. The figure of subsidy expenditure has been rising steadily over the study period – from around ₹28 thousand crore in 2005 to around ₹145 thousand crore in 2023, representing more than five times the amount spent during the same period. The growth has been particularly strong since 2018, due to more support provided by the government for agriculture, such as subsidized electricity for irrigation, the cost of electricity procurement has risen and the demand for electricity in agriculture has grown. These subsidies have helped irrigators offset irrigation costs, boost farmers' incomes, and have

helped to overuse groundwater resources because they have reduced the cost of groundwater extraction. Furthermore, the increasing subsidy costs create a fiscal challenge for the state governments and electricity distribution companies, underscoring the importance of implementing policy changes that can encourage the adoption of energy-efficient irrigation methods, fair pricing of electricity and groundwater resources, and sustainable groundwater management.

4.1 Fiscal and Environmental Costs

The total financial burden of electricity subsidies in India's state governments was approximately INR 1.45 lakh crore (about USD \$17.5 billion) in 2022-23, and contributed significantly to the overall fiscal deficit for each state. Notably, some states, such as Punjab, Andhra Pradesh and Tamil Nadu, spend approximately 10-15% of their total revenue expenditures on agriculture electricity alone (Gulati and Banerjee, 2015).

On the other hand, the cost-free power supply also has negative environmental impacts: it removes the one price signal that would otherwise provide discipline to farmers in the way they extract from groundwater. A farmer with an increased depth to water in the aquifers will normally respond to higher pumping costs by reducing the amount of water being pumped. However, if the farmer is provided free power or a flat rate for the same, he or she would not receive that price signal. There is compelling evidence to support the idea that providing price signals matter at the margins: by doing a natural experiment, it can be shown that where power was metered and charged for, groundwater tables stabilised or increased, but that the opposite result occurred in the states where there was no price signal (Shah et al., 2008).

4.2 Political Economy of Subsidy Persistence

What are the reasons behind the continuous existence of subsidies which are both inefficient, wasteful of funds and damaging to the economy? The political economy literature explains this phenomenon as a well-documented trend: once a program like an agricultural subsidy is put in place, the groups receiving benefits from it will become organised, concentrated, have powerful lobbying forces and as a result will also exert considerable amounts of political pressure. In addition, the costs associated with these programs will be spread out over many tax payment recipients that do not have any political organisation, nor will they be well represented in the immediate election process (Olson, 1965). Examples from Indian state elections (Punjab-2022, Andhra Pradesh-2019, Tamil Nadu-2021) clearly demonstrate how free agricultural electricity to farmers can be used to win or lose elections. The average farmer in northern India does not have a substantial amount of irrigated land; in fact, a sizable portion of them will be small farmers whose only means of supporting themselves is through the provision of free agricultural electric power supply as a true source of income. Therefore, the presence of free agricultural electricity provides farmers with both productive distortion at a time when reforming those types of programs could create political difficulties and at a time when it creates an economic problem.

5. Pricing Mechanisms for Correcting Groundwater Market Failure

Economic theory tells us that there are many tools available to use for fixing open-access problems and also ways to fix negative externalities. In this paper we will assess 4 different types of tools in terms of how they will work in India, looking primarily at their ability to create an

efficient outcome, be fair to producers and consumers as well as whether the tool could logistically be used by any government agency.

5.1 Direct Pigouvian Groundwater Taxation

A solution that is theoretically clean to resolve negative externalities is a Pigovian tax. The purpose of this tax is to impose an additional cost on each cubic meter of groundwater that is extracted. This will also raise the private cost of extracting groundwater equal to the social cost, allowing the equilibrium of groundwater to shift from Q^* (market) to Q^{**} (social optimum). To implement a Pigovian tax on the extraction of groundwater, all extractions must be metered volumetrically and charged at a rate equal to the external costs associated with groundwater depletion.

Efficiency Gains: Modeling by the World Bank (2010) indicates that the introduction of a Pigovian charging scheme for groundwater at an estimated charge of between INR 1-3/m³ in severely over-exploited zones will result in a reduction in agricultural extraction of 15-25%, without any significant losses in crop production. Farmers will adjust their extraction practices as they adopt more water-saving methods of irrigation (drip or sprinkler systems), and crop types that plant less on a unit area basis, in conjunction with implementing the charge. Kumar et al. (2021) estimated that groundwater use efficiency in Gujarat could be expected to improve by approximately 18-22% at a reasonable charge of INR 0.5/m³, as technology was adopted to assist farm operations with improved groundwater management.

Limitations: There are two major practical hurdles in India to direct taxation of ground water: (a) The lack of volumetric metering for the estimated 30-40 million irrigation wells (the monitoring and enforcement infrastructure needed is not feasible to scale up in the near term); and (b) the political opposition of farm lobby groups to any direct water charge, which would be subject to

constitutional challenges, since water is recognized as a fundamental right in Indian law. Direct Pigouvian taxes on water are not a short-term policy lever, due to the structural limitations.

5.2 Electricity Tariff Reform: Increasing Block Tariffs

The potential for directly taxing water is constrained by the metering requirements and the need to keep electricity prices low for water treatment and distribution, making the reform of agricultural electricity tariffs the most operationally viable pricing correction. Under the Increasing Block Tariff (IBT) there are progressive increases in the price per kWh based on consumption blocks defined by the decreasing intensity of pumping; there is thus an implicit water price signal. Given approximate constancy of the head (hydraulic pressure), the IBT can be used as a realistic indicator of water extracted from the ground, and is therefore a useful proxy for water taxation without requiring volumetric water meters.

The most instructive evidence comes from Gujarat's Jyotigram Yojana, which was launched in 2003. The scheme effectively increased the cost to the large irrigators and enhanced supply quality for all by separating the agricultural feeder from the domestic feeder and giving the farmers 8 hours of reliable, metered, 3-phase power (whereas they were metered with 1 phase and unreliable power for the past 24 hours). Shah et al. (2008) have reported that the groundwater table recovery has been observed in the districts covered by Jyotigram and it was found that the groundwater table level stabilized in 73% of the monitored wells in five years with statistical significance. The scheme has also been beneficial for the finances of Distribution Company (DISCOM) as it has reduced the subsidy burden by around 25%.

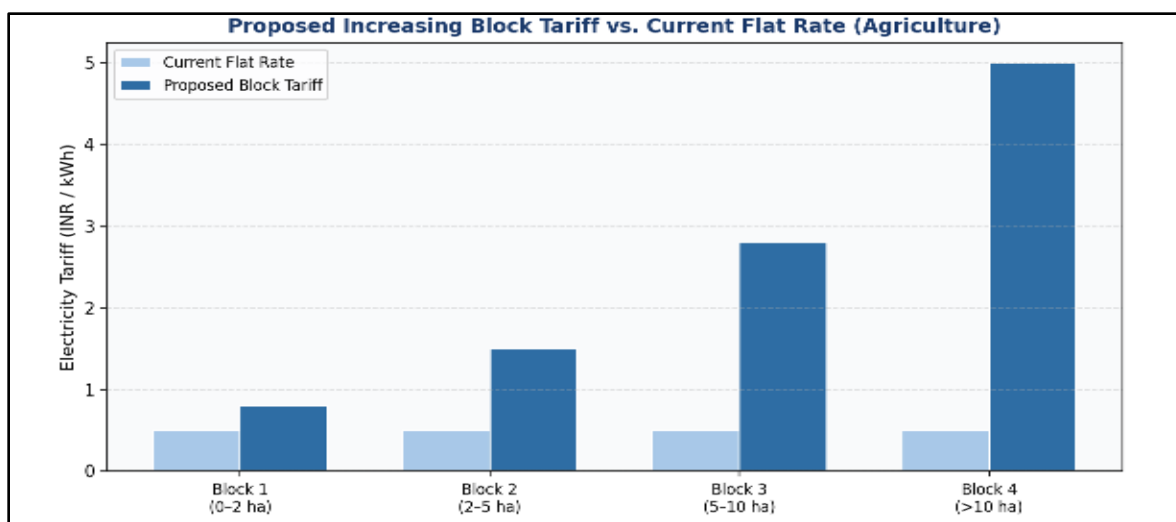


Figure 7: Proposed Increasing Block Tariff Structure vs. Current Flat Rate. Block 1 (small landholders < 2 ha) receives near-current rates to protect marginal farmers; large irrigators (> 10 ha) face market-reflective pricing.

Sources: Author's construction based on Gulati & Banerjee (2015) and Ministry of Power Tariff Guidelines (2022).

Block calibration is an important aspect of the technology of IBTs when it comes to design of equity. The small farmers (holding less than 2 hectares) rely on groundwater for their survival and do not withstand high tariff hikes without income support. However, an IBT structure must design an affordable rate of the 1st block for smaller farmers (preferably at the same rate as smaller farmer with government subsidy continuing) and sharply increase the rate for larger farmers who extract disproportionate quantities. (Gulati and Banerjee, 2015).

5.3 Tradable Groundwater Rights Markets

An even more drastic market correction is the establishment of well-defined, quantifiable and tradable groundwater rights. In this model, each aquifer is periodically evaluated for sustainable yield, and a set of equal sustainable yield extraction rights are issued annually, which

may be traded between different users and water allocated according to price discovery to the highest-value use (Dinar and Subramanian, 1997).

The potential benefits are substantial in theory: trade makes water go to where it will yield the greatest marginal benefit, avoiding the "dead weight loss" that occurs when water is wasted on low value crops. Partial evidence of efficiency gains can be seen in Australia's Murray-Darling Basin experience and Chile's water markets under the 1981 Water Code, although in both instances distributional and ecological issues arose (Grafton et al., 2011).

The situation is not as favorable in India as it is in the USA, as there are many obstacles to the tradability of rights in India. The property rights foundation is missing: there is no national or state groundwater law that sets out to define and register volumetric rights to groundwater extraction. The proposed "Model Groundwater (Sustainable Management) Bill (2017)" by the Ministry of Jal Shakti has yet to be implemented in many states. Further, the technical difficulty in defining a single 'unit of right' is complicated by the fact that aquifer heterogeneity means that a right to extract 1,000 m³ in Punjab may be different to a right to extract 1,000 m³ in Karnataka, depending on the vastly different recharge rates and aquifer characteristics in the two regions connectivity (Singh, 2004).

5.4 Community-Based Quota Systems

A fourth method is based on the concept of community-determined groundwater quotas, as opposed to market pricing, and is based on Ostrom's (1990) framework. This is a model in which an Aquifer Water Users' Association (AWUA) (at the aquifer or sub-aquifer level) collectively sets extraction quotas, monitors compliance, and applies graduated sanctions to those who violate the agreement. The state's responsibilities are to give legal status to the

AWUA and to facilitate the exchange of hydrological data and/or to settle conflicts between communities.

India has some success stories: Pani Panchayat model in Orissa/Odisha where the villages jointly managed the surface water allocation with minimal external involvement (Patel, 2010); Arvari River Parliament (Jal Biradari) in Rajasthan which has restored the seasonal river through community governance by managing the recharge and reducing the extraction; Participatory Groundwater Management (PGWM) pockets in Andhra Pradesh and Gujarat under the National Water Mission (Ministry of Jal Shakti, 2023).

But community-based approaches are best suited at small scales involving homogeneous user groups. Aquifers which span administrative boundaries and are large and deep in extent, such as the Ganga alluvial plain, serve millions of users whose interests are far too varied to be managed by communities. These measures can thus best be considered as complementary to, not a replacement for, price and regulatory measures (Narain, 2014).

5.5 Comparative Evaluation

Mechanism	Economic Efficiency	Equity Outcome	Administrative Feasibility	Political Feasibility
Pigouvian Water Tax	High	Regressive (needs rebate)	Low (metering gap)	Very Low
Electricity IBT	Medium-High	Manageable (block design)	Medium (smart meters)	Medium
Tradable Rights	High (long-	Risk of	Low (legal reform)	Low

Market	run)	concentration		
Community Quota	Medium	Equitable	Medium (small scale)	High

Table 1: Comparative Evaluation of Pricing Mechanisms for Groundwater Management in India. Sources: Author's synthesis based on Gulati & Banerjee (2015), World Bank (2010), Dinar & Subramanian (1997), Ostrom (1990), and Kumar et al. (2021).

6. Policy Implications

The above analysis presents a series of policy recommendations along three complementary strands: (i) price reform to introduce internalization of the externalities; (ii) institutional reform to create rights and governance; and (iii) technology and information investment to facilitate market and regulatory functions.

6.1 Phase Out Flat-Rate Electricity Subsidies, Introduce IBT

The most important action that could be taken by policy now is to move to an Increasing Block Tariff structure for agricultural electricity. When executed with supply quality enhancements and proper farmer preparation, this is very much possible without compromising on the productivity of the agricultural supplies, as witnessed in Gujarat Jyotigram (Shah et al., 2008). We recommend:

- (a) Block 1 (holdings < 2 ha): continue current subsidised rate or INR 0.5-1/kWh with the targeted cash transfer compensation done through PM-KISAN DBT to keep the welfare of the marginal farmers unaffected.
- (b) Block 2 (2–5 ha): INR 1.5–2.5/kWh, reflecting partial cost recovery.

- (c) Block 3 (5–10 ha): INR 2.5–3.5/kWh, approaching cost-reflective rates.
- (d) Block 4 (> 10 ha): Subsidy removed and price to large common commercial irrigators at LRMC (INR 5-8/kWh).

Revenue should be recycled for aquifer recharge systems, expand micro-irrigation and support smallholder incomes, while being segregated and allocated within a 'double dividend' argument that is politically viable (Pigou, 1920; Gulati and Banerjee, 2015).

6.2 Enact a National Groundwater Governance Framework

The current absence of a legally binding framework for the governance of groundwater in India is a crisis, as India needs a framework that can be implemented across the country. A good beginning is provided with the Model Groundwater (Sustainable Management) Bill, 2017. The following should be included:

- (a) All extraction wells over a certain diameter should be registered and and extraction should be metered either by volume or by proxy (proxy metering is the use of a water level meter to measure with reference to the water level of the extraction well).
- (b) Legally established Aquifer Water Users' Associations (AWUAs) at the aquifer-unit level and representation of user-classes.
- (c) Clarification of 'prior-appropriation' water-use rights for water-intensive crops in order to facilitate future markets for water-use rights.
- (d) Compulsory Environmental Aquifer Impact Assessment (EAIA) for any new irrigation scheme on a large scale or urban water extraction project.

6.3 Invest in Monitoring Infrastructure and Data Transparency

There are currently 22,000 monitoring wells in the CGWB network, which is completely insufficient to manage a resource used by 30–40 million wells. The density of one calibrated monitoring well per 3 km² should be achieved across over-exploited areas within India in the next ten years. Groundwater level sensors with low cost- and easy-to-track IoT technology should be expanded across the country (Ministry of Jal Shakti, 2023) and linked to state-level dashboards as piloted under the National Aquifer Mapping and Management Programme (NAQUIM). Community monitoring of extraction and water table data would be possible and information asymmetry would be diminished if extraction is published in open data format at the aquifer level.

6.4 Crop Diversification and MSP Reform

If we are still providing income for growing paddy through a Water Scarcity Management Plan (MSP) in the states of Punjab and Haryana, the pricing mechanisms cannot operate in isolation. Expanding MSP coverage to include additional water-efficient crops (i.e. millets, pulses and oilseeds) will help to establish a procurement infrastructure in areas that are not traditionally growing such crops and reduce the incentive to grow paddy in regions that should not grow paddy due to climate limitations. The Haryana Sahi Fasal Yojana (2018-2022) provided farmers with INR 7,000 per acre to transition from paddy to other crops with low water-intensity consumption and reduced paddy acreage by approximately 90,000 hectares, saving approximately 1.5 BCM of groundwater each year. This is clear evidence to support the concept (CGWB 2022; Narain 2014).

6.5 Promote Micro-Irrigation with Water Credit

In his research work, Kumar and his co-authors (2021) provide evidence that agriculture can decrease its demand for water by an estimated 30-50% through the use of demand side efficiency measures such as drip and sprinkler irrigation. The Government of India's Pradhan Mantri Krishi Sinchayee Yojana (PMKSY) program "Har Khet Ko Pani, More Crop Per Drop" is currently subsidizing the use of micro-irrigation; however, only about 20 percent of the target area has adopted micro-irrigation due to high initial capital costs, lack of credit availability, and gaps in technical assistance. To improve micro-irrigation use, we recommend linking tradable groundwater "water credits," which represent the volume of water saved by an investment in efficient irrigation systems, to the adoption of micro-irrigation systems, thus providing a financial incentive for micro-irrigation use that is not dependent on the rate of water tariffs.

6.6 Equity-Sensitive Phasing and Transition Support

Reforming a pricing structure which results in increasing the price of water or electricity will disproportionately impact smallholders who do not have the capital resources needed to invest in improving efficiencies, and operate at thin margins in their farming operations. Thus, a reform that is socially legitimate must include: (a) the provision of direct cash transfers to small and marginal farmers to compensate for the higher energy costs during this period of transition; (b) subsidised access to micro-irrigation equipment for all smallholders through credit lines supported by NABARD; (c) an expansion of crop insurance to limit the risk to farmers income during this transitional period; (d) grievance redress mechanisms for farmers at the block level. Examples from Spain, Morocco, and Mexico of international water pricing reform demonstrate that reforms which do not take equity into consideration can result in a political reversal;

however, those that are sensitive to the issues of equity will result in a durable reform (OECD, 2010).

7. Limitations and Future Research

There are many limitations associated with paper's conclusions. In addition to several of the conclusions being based on estimates made from modeling studies and hedonic pricing which were based on assumptions regarding aquifer connectivity, spatial spillover effects, and farmers' behavioral responses; the great variability between India's aquifer systems lends itself to variability amongst any one MEZ for either system. Thus, aquifer specific hydrologic models coupled with choice experiment surveys (measuring willingness to pay and willingness to accept) would greatly enhance the empirical basis for establishing tariff rates on the use of groundwater resources.

Further compounding the ambiguity in this paper's conclusions is the fact that it does not account for the entire general equilibrium effect of groundwater pricing on food prices, wages in rural areas and the outmigration of people from rural to urban areas. Agricultural prices may potentially be impacted due to increased costs of agricultural water, even if the increase in agricultural water costs is offset by making targeted resource transfers to the producers of agricultural commodities, thereby impacting the prices of agricultural commodities produced in the market (e.g., grain market, vegetable market, dairy market, etc.), which affects the price of food is likely to the urban poor given the substantial number of individuals living below the poverty line in India. Thus, more research using computable general equilibrium (CGE) models to evaluate IBT scenarios would certainly add value to the existing literature.

The authors of this paper have mainly focused on the issue of crop pricing through market failures and remedial measures; however, there are many other non-pricing policy instruments available for addressing the groundwater depletion problems. Additional attention should also be given to investigating complementary types of policies, such as legislation that includes withdrawal limits from registration of well (the Wells Act), mandatory crop rotation requirements, and regulated recharge of wells (i.e. artificial recharge).

In order to formulate a more complete framework for implementing the above policy instruments, future research should give more emphasis to:

- (i) conducting randomised controlled trials (RCT) or quasi-experimental evaluations at the aquifer level regarding IBT and community quota pilot programs.
- (ii) satellite-based monitoring of groundwater storage using GRACE-FO data at district level.
- (iii) conducting political economy analyses of subsidy reform sequencing to assess coalition-building dynamics.
- (iv) conducting comparative studies of pricing reform success among similar developing countries such as Iran, Pakistan, Bangladesh, and Morocco.

8. Conclusion

The government policy and market institutions in India have failed to establish a pricing structure for an ecologically interconnected resource (namely groundwater) using that resource's social opportunity cost. The analysis presented in this paper indicates that, to date, groundwater depletion is a multi-dimensional market failing: negative externalities represent a significant cost on third-party actors (e.g., farmers), as well as future generations; the commons problem creates

an individually rational but collectively destructive practice of extraction; and lack of information symmetry creates significant challenges to efficient planning. All of these failures are further exacerbated by government-supplied electricity subsidies that exceed INR 1.45 lakh crore annually, which creates a near-zero marginal cost of pumping and effectively accelerate the tragedy of the commons.

The most economically efficient methods to correct the failures associated with groundwater markets are through price mechanisms (e.g., Pigovian taxes). The fundamental mistake in this market is the difference between the marginal private and social costs of groundwater extraction. The optimal correction to this failure is to charge a tax or use a price equal to the marginal external cost of groundwater use, in order to restore efficiency to the market for groundwater resources. In India, direct groundwater taxes are administratively infeasible in the immediate term, while the use of increasing block tariffs (IBTs) for the pricing of electricity, as demonstrated through the Jyotigram model of Gujarat, represents the best short-term mechanism available to India for addressing the failures in its groundwater markets. Tradable rights markets are ultimately a long-term goal that will involve the need for reforming both legally and institutionally. Additionally, community-based quota systems will serve as an additional form of governance for smaller, limited aquifer systems.

However, unless there are simultaneous efforts underway with regard to crop incentives (namely MSP reform), increased monitoring through expanded networks by CGWB, and providing equity protection through targeted cash transfers to smallholders, then no pricing mechanism will be successful. Successful subsidy reform is tied to the ability to communicate efficiency gains clearly and credibly compensate those people that will meet loss due to the changes. The CGWB (2022) data show that Indian policymakers are working with a very short

time period as more than 25% of total assessment units show that they are under high levels of stress without taking corrective action, leading to regional food security crises within two decades. The mechanisms for correcting this market failure are available, but it is only through institutional commitment that they can be implemented.

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